Development and evaluation of deep learning models for detecting and classifying various bone tumours in full-field limb radiographs using automated object detection models

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Aims

We aim to develop a fully automated deep-learning model to detect and classify benign/malignant bone tumours in full-field limb radiographs using an object detection model. The secondary aim is to identify differences in classification characteristics between the developed automated model, three orthopaedic oncologists, and three general orthopaedic surgeons.

Methods

This retrospective analysis included 642 limb bone tumours with 40 diagnoses confirmed pathologically from three institutions (378 benign, 264 malignant including intermediate types). We employed end-to-end object Detection with transformers with Improved deNoising anchOr boxes (DINO) and You Only Look Once (YOLO) models. We performed five-fold cross validation on the collected radiographs, using the training data to train the models, validation data to optimize the models' parameters, and independent test data for final performance evaluation. Firstly, we confirmed DINO achieves a higher detection rate than YOLO. Secondly, we compared the classification performance of DINO with those of doctors, using various metrics such as accuracy, sensitivity, specificity, precision, and F-measure.

Results

The DINO model achieved a higher mean tumour detection rate (85.7% (95% CI 81.5 to 89.8)) than the YOLO model (80.1% (95% CI 77.2 to 82.9)). For the evaluation of classification performance, we used 113 cases that DINO detected out of 128 randomly selected cases as the evaluation test set. The accuracy and sensitivity of the DINO model, as a superior model, were significantly higher than those of general orthopaedic surgeons. The DINO model correctly classified 78.6% (22 out of 28 cases) of the challenging cases that two or more doctors misclassified. However, DINO's diagnostic errors primarily occurred with tumours that were diagnostically challenging for orthopaedic oncologists or present in unusual sites.

Bone & Joint Research

Conclusion

The DINO model automatically detects bone tumours better than the YOLO model, and may assist doctors in detecting tumours and classifying malignant/benign bone tumours in clinical practice.

Article focus

- Developing a deep learning model to detect and classify bone tumours in limb radiographs automatically.
- Comparing transformer-based and convolutional neural network (CNN)-based object detection models in bone tumour detection.
- Evaluating the model's diagnostic accuracy through comparison with three orthopaedic oncologists and three general orthopaedic surgeons.

Key messages

- The transformer-based Detection with transformers with Improved deNoising anchOr boxes (DINO) model demonstrated superior performance over the CNN-based You Only Look Once (YOLO) model in bone tumour detection on limb radiographs.
- The model achieved diagnostic accuracy exceeding all general orthopaedic surgeons and some orthopaedic oncologists in tumour classification.
- This AI system has demonstrated potential as a clinical decision support tool to reduce misdiagnosis of bone malignancies.

Strengths and limitations

- The study demonstrated the superiority of the transformer-based DINO model in bone tumour detection and classification.
- This study validated the results using state-of-the-art object detection models on the same bone tumour dataset. Furthermore, the proposed method demonstrated a statistically significant superiority compared to the other approach.
- The study was limited by its relatively small sample size of 642 cases and the exclusion of non-tumour conditions such as degenerative diseases.

Introduction

Early and precise identification of bone tumours markedly impacts the prognostic outcome. Osteosarcoma, the most common primary malignant bone tumour, has a five-year survival rate of 60% to 80% in patients without pulmonary metastasis at initial diagnosis.[1–3] Although osteosarcoma and Ewing's sarcoma are treated with chemotherapy,[4] surgical resection is essential for the curative treatment of malignant bone tumours.[5] The prognosis of patients who are difficult to resect is inferior.[6]

Orthopaedic surgeons must not overlook a malignant tumour in a radiograph. Radiography is initially performed as a simple screening test, and can provide information on the presence or absence of tumours and the classification of benign or malignant.[7,8] However, it is difficult for general orthopaedic surgeons to classify bone tumours because there are many types,[9] and primary malignant bone tumours are rare, accounting for only 0.2% of all tumours.[10–12] In addition, the rarity and diversity of these make it difficult for general orthopaedic surgeons to gain sufficient experience.

Automated bone tumour diagnostic models using AI have been developed to classify benign and malignant bone tumours using radiographs.[13–16] In particular, several papers have reported that their methods outperformed the accuracy of radiologists, general orthopaedic surgeons, and orthopaedic oncologists.[13,14] However, in many studies on deep learning models for bone tumours, the images used were limited to specific radiographs (e.g. only knee or hip) or manually cropped images of the tumours.[13,14,16,17] To solve these limitations, AI-based object detection models, such as Detectron2 and You Only Look Once (YOLO) models, were used for automatically detecting and classifying bone tumours on full-field radiographs of various body parts.[18,19] The YOLO model was promising because of its higher detection accuracy than other models such as Detectron2. However, even the YOLO model did not perform well in classifying malignant tumours. This performance limitation arises from its model architecture; specifically, YOLO is based on a convolutional neural network (CNN), which is a popular AI model for image classification and is also widely used for medical images.[20–26] CNN extracts local features and aggregates them to make its classification or detection without evaluating the relationship among distant local parts.[27] In contrast, the bone tumour detection and classification task requires more contextual (i.e. non-local) information, which is available by observing and evaluating the relationship among distant parts in full-field radiograph images.

To overcome these limitations, we propose a new automatic bone tumour detection and classification model. The model is based on end-to-end object Detection with transformers with Improved deNoising anchOr boxes (DINO),[28] which is a state-of-the-art object detection model. DINO employs a Transformer architecture. Unlike CNN, the Transformer has an attention mechanism that evaluates the relationship among distant image parts,[27,29] and may be able to learn whole bone shapes, such as those in limb radiographs. Since tumour locations and types are often dependent on bone types and their whole shapes, the Transformer architecture is expected to be useful in detecting a bone tumour and classifying whether the bone tumour is malignant or benign.

In this paper, using radiographs of patients with various bone tumours from three institutions with bone tumour departments, we first show that our DINO-based model can detect bone tumours more accurately than YOLO. We then show that our model can classify bone tumours with accuracy that is comparable to or even higher than orthopaedic oncologists or general orthopaedic surgeons. These positive results prove that both tumour detection and classification tasks need wider views to capture non-local relationships in each full-field radiograph image.

Methods

Dataset

The Institutional Review Board of Kyushu University Hospital (No.22316-00, No.22089-00) approved this study. Cases with a radiograph of the limb with a visible bone tumour at the initial visit and with a definitive pathological diagnosis were included in our dataset (cases with multiple bone tumours were excluded). The cases were collected from three tertiary referral centres that have departments specializing in bone tumour treatment because these cases included a wide range of bone tumours, including both common and rare types. The patient population encompasses various age groups, tumour locations, and disease severities, ensuring the representativeness of the general population of bone tumour patients requiring treatment in Japan. Table I shows the details of the dataset. We divided this dataset into five subsets for five-fold cross-validation in the later experiment. This table also details the evaluation test set; we compare the classification performance of an object detection model with those of orthopaedic oncologists and general orthopaedic surgeons.

This retrospective study included 642 cases that underwent biopsy or surgical resection from three hospitals, comprising 40 bone tumour types (19 benign, four intermediate, and 17 malignant) based on pathological diagnoses: Kyushu University Hospital (Fukuoka, Japan), Kyushu Rosai Hospital (Kitakyushu, Japan), and the National Hospital Organization Kyushu Cancer Center (Fukuoka, Japan). The pathological diagnosis was established by experienced musculoskeletal pathologists (see Acknowledgements) following WHO classification criteria.[9] Only cases with conclusive pathological diagnoses were included in the study. Kyushu University Hospital provided 432 cases (252 benign, 38 intermediate, and 142 malignant) between February 2006 and December 2022. The National Hospital Organization Kyushu Cancer Center provided 64 cases (22 benign, 20 intermediate, and 22 malignant) from January 2006 to November 2021. Kyushu Rosai Hospital provided 146 cases (104 benign, four intermediate, 38 malignant) between April 2016 and December 2022. The cases were categorized into 378 benign and 264 malignant. The intermediate types were included in the malignant types because a biopsy is clinically required. We obtained 1,129 radiographs from these cases. The radiographs included the referring hospital's radiographs, radiographs at the initial visit, and preoperative radiographs.

In this study, to detect the tumour's location and accurately classify the tumour, it was necessary to correctly annotate each identified lesion by labelling a bounding box (a rectangular frame surrounding the entire tumour) and assigning the appropriate benign or malignant classification. First, one orthopaedic surgeon (MY, 10 years of experience) labelled the bounding boxes on the radiographs using LabelMe (an open-source image-labelling programme based on Python).[30] Second, two experienced orthopaedic oncologists (TH (see Acknowledgements) and ME; 13 and 23 years of experience, respectively) checked and revised the bounding boxes. These annotators were independent of the orthopaedic oncologists (AN and SI (see Acknowledgements), NY) and general orthopaedic surgeons (ET, RH, KM (see Acknowledgements for all)) who later participated in the performance comparison study between DINO and doctors. Third, for each identified tumour, the binary ground truth (benign or malignant) was assigned based on pathological diagnoses of specimens obtained through surgical resection or biopsy.

Object detection models and training procedure

Figure 1 illustrates this method's overview. As preprocessing, the input images were resized to 512 by 512 pixels, and the left-side images were flipped to the right, except for the hip images. The images were then fed into the object detection models. The models automatically detected tumours as bounding boxes and classified them as benign or malignant. The DINO model[28] and the YOLO-v8x model[31] were used to detect and classify tumours because the models are object detection models with excellent performance. Both DINO and YOLO models are open-source frameworks that are freely available for implementation via GitHub. These models were selected to address two clinical challenges: the difficulty orthopaedic surgeons face in detecting subtle bone tumours on radiographs, and the limited experience many general orthopaedic surgeons have in differentiating between benign and malignant tumours. The DINO model is based on the so-called Transformer,[29] which employs the attention mechanism. Roughly speaking, the attention mechanism can calculate (adaptively weighted) spatial correlation over the whole input image and thus captures its global image features.[32] Consequently, the DINO model has a potential of recognizing age-related bone structure variations and tumour predilection sites by analyzing relationships between local tumour features and distant anatomical landmarks. In contrast, the YOLO model is based on a CNN, which employs layered convolution operations to aggregate local information and therefore is suitable for extracting local features.[27] "v8x" is the recent version of YOLO models used in related studies and other medical field studies.[19,33–35] By training these models on full-field radiographs, we developed systems that align with clinical practice to assist doctors in making faster and more accurate diagnostic decisions. For training and evaluating these object detection models, we implemented a five-fold cross-validation. The 642 cases in the dataset were randomly divided into five subsets (Fold0-Fold4), ensuring no overlap across the different subsets. Next, each subset was used as a test dataset, and the cases in the other four subsets were randomly divided into validation (20%) and training (80%) sets. Each test dataset included only radiographs obtained at the initial visit. For each fold of five-fold cross-validation, we trained the models using training data, optimized the models' hyper-parameters (especially, training epochs, that is, the timing to stop training) using validation data, and performed the final evaluation using test data. Through this cross-validation process, performance results were obtained from five different model evaluations. The details of the number of cases and images used are shown in Supplementary Figure a. We used data augmentation (cropping and rotation) during training to compensate for the limited data and variations in bone positioning and angular differences during radiological imaging. Examples of image rotation and cropping for data augmentation are shown in Supplementary Figure b. For technical details on object detection models, see Supplementary Methods.[30,31,36–38]

Table I. Patients' characteristics.

Characteristic	All cases	Evaluation test set
Benign	378	62
Mean age, yrs (SD)	28 (19)	
Sex (female), n (%)	168 (44)	
Tumour types, n (%)		
Enchondroma	98 (25.9)	16
Osteochondroma	60 (15.9)	7
Simple bone cyst	47 (12.4)	10
Osteoid osteoma	40 (10.6)	6
Fibrous dysplasia	31 (8.2)	9
Non-ossifying fibroma	22 (5.8)	4
Chondroblastoma	19 (5.0)	5
Intraosseous ganglion	9 (2.4)	1
Bizarre parosteal osteochondromatous proliferation	8 (2.1)	
Osteofibrous dysplasia	7 (1.9)	2
Langerhans cell histiocytosis	7 (1.9)	2
Lipoma of bone	6 (1.6)	
Periosteal chondroma	6 (1.6)	
Subungual exostosis	5 (1.3)	
Aneurysmal bone cyst	5 (1.3)	
Chondromyxoid fibroma	3 (0.8)	
Haemangioma of bone	2 (0.5)	
Epidermoid cyst	2 (0.5)	
Glomus tumour	1 (0.3)	
Intermediate	62	15
Mean age, yrs (SD)	38 (19)	
Sex (female), n (%)	34 (55)	
Tumour types, n (%)		
Giant cell tumour of bone	56 (90.3)	12
Osteoblastoma	3 (4.8)	1
Desmoplastic fibroma	2 (3.2)	1
Fibrocartilaginous mesenchymoma	1 (1.6)	1
Malignant	202	36
Mean age, yrs (SD)	45 (24)	
Sex (female), n (%)	87 (42)	
Tumour types, n (%)		
Osteosarcoma	70 (34.7)	13
Metastatic bone tumour	60 (29.7)	12
Chondrosarcoma	20 (9.9)	5
Malignant lymphoma	11 (5.4)	1
Malignant giant cell tumour of bone	7 (3.5)	1
Leiomyosarcoma of bone	5 (2.5)	1
Ewing's sarcoma	5 (2.5)	2
Dedifferentiated chondrosarcoma	5 (2.5)	1
Plasmacytoma of bone	5 (2.5)	
Mesenchymal chondrosarcoma	3 (1.5)	
Periosteal chondrosarcoma	3 (1.5)	
Adamantinoma	2 (1.0)	
Undifferentiated pleomorphic sarcoma	2 (1.0)	
Malignant myoepithelioma	1 (0.5)	
Epithelioid hemangioendothelioma	1 (0.5)	
Clear cell chondrosarcoma	1 (0.5)	
Undifferentiated small round cell sarcoma	1 (0.5)	

(*Continued*)

Evaluation methods of differences in performance between DINO and YOLO

The detection performance of the DINO model was compared with that of the YOLO-v8x to confirm its superiority to the YOLO-v8x. We used several metrics to evaluate the tumour detection performance: overall detection rate, those for benign and malignant tumours, and mean Intersection over Union (IoU). For each input image, the models output candidate bounding boxes, their classes (benign or malignant), and their corresponding confidence scores. We selected the bounding box with the highest confidence score as the final estimation result of the models. The detection rate was then calculated as the percentage of cases that were successfully detected in the test dataset. We considered the detection successful if the IoU exceeded a threshold of 0.1. This threshold was used because a 10% overlap with the actual region is sufficient for orthopaedic surgeons to localize the proper area.

Evaluation methods of differences in performance between DINO and doctors

To quantify the clinical efficacy of the DINO model, we compared its tumour classification performance with those of three experienced orthopaedic oncologists (Dr. SA, Dr. SB, and Dr. SC; 17, 12, and 18 years of experience, respectively) and three general orthopaedic surgeons (Dr. GA, Dr.GB, and Dr. GC; 10 years of experience).

We used several metrics to evaluate tumour classification performance: accuracy, sensitivity, specificity, precision, and F-measure, which were calculated using scikit-learn.[39] Note that these metrics were assessed using the evaluation test set, which included the successful detection cases of the superior model's Fold1-test set. This set was used because the superior model achieved a slightly better tumour detection

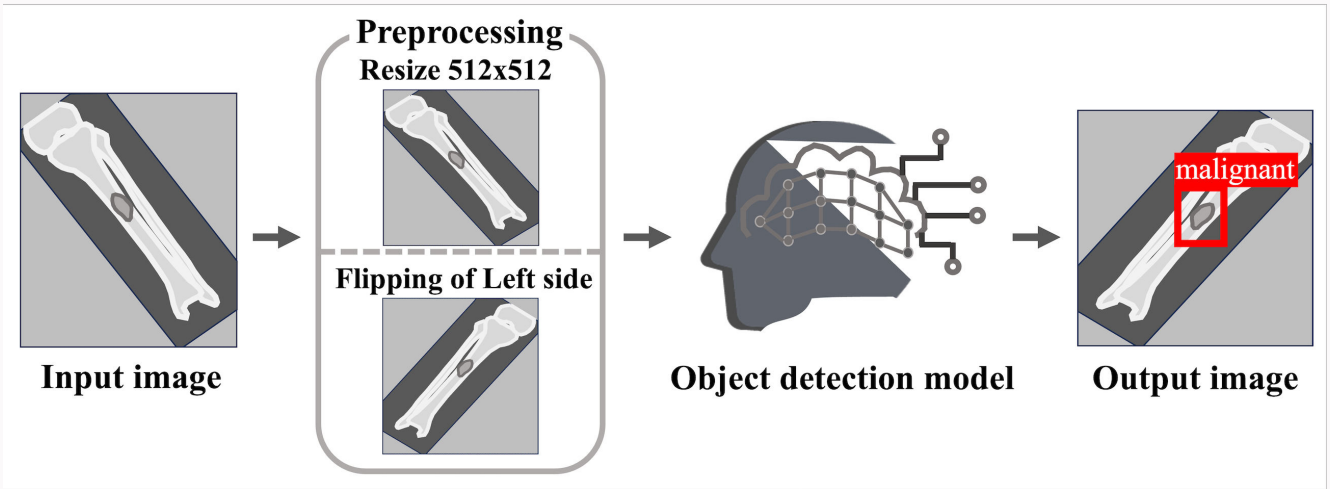

Fig. 1
Overview of object detection models.

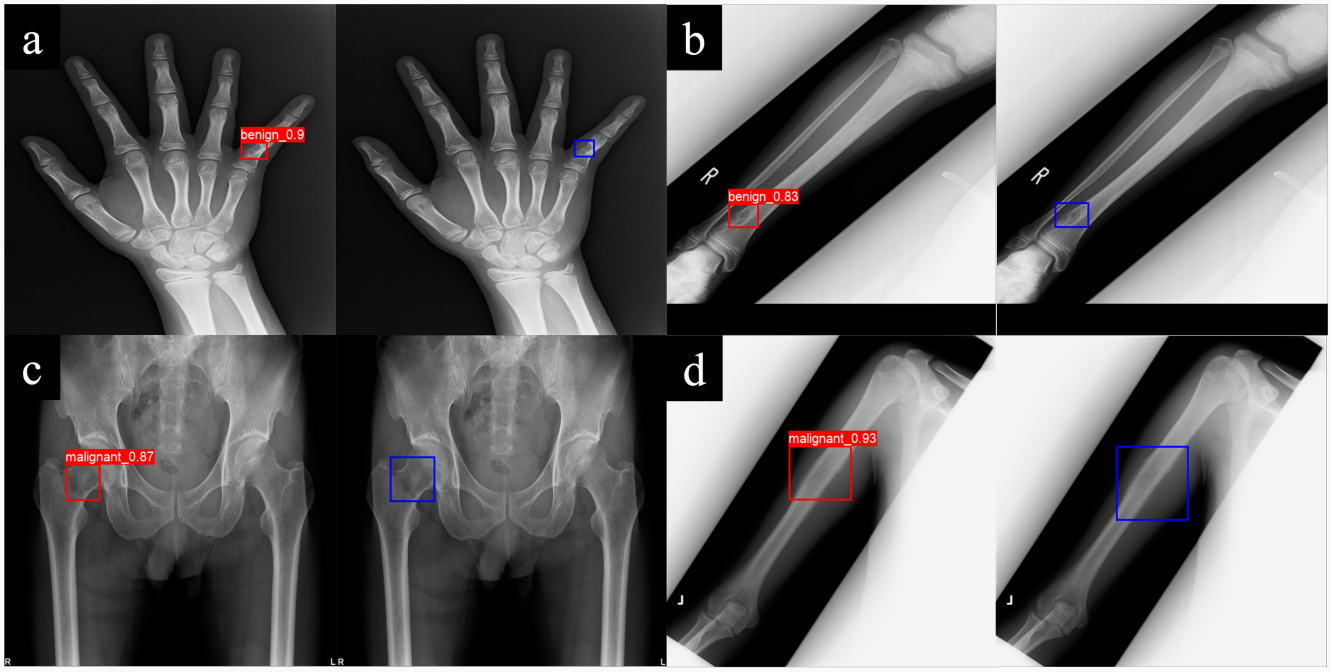

Fig. 2
Correct predictions by the Detection with transformers with Improved deNoising anchOr boxes (DINO) model. The blue bounding box is a ground-truth bounding box, and the red bounding box is a predicted bounding box. a) Periosteal chondroma case (79-year-old female), b) non-ossifying fibroma case (12-year-old male), c) giant cell tumour of bone case (53-year-old male), and d) Ewing's sarcoma case (17-year-old male). All radiographs were anteroposterior views obtained at the initial visit.

rate among the five subsets. Accuracy is calculated as the number of correct classifications divided by the number of evaluated cases. Sensitivity is the rate of tumours predicted to be malignant out of the total number of malignant tumours. Specificity is the rate of tumours predicted to be benign out of the total number of benign tumours. Precision is the rate of tumours that are actually malignant out of the tumours predicted to be malignant. F-measure is the harmonic mean between precision and sensitivity.

Statistical analysis

The statistical tests of the evaluation metric scores were performed using the JMP ver.17 (SAS Institute, USA). A Mann-Whitney U test or paired *t*-test were used to analyze the difference in continuous variables of tumour sizes and the models' detection rates. The McNemar test was used to compare the classification performance. The level of significance was set at $p < 0.05$.

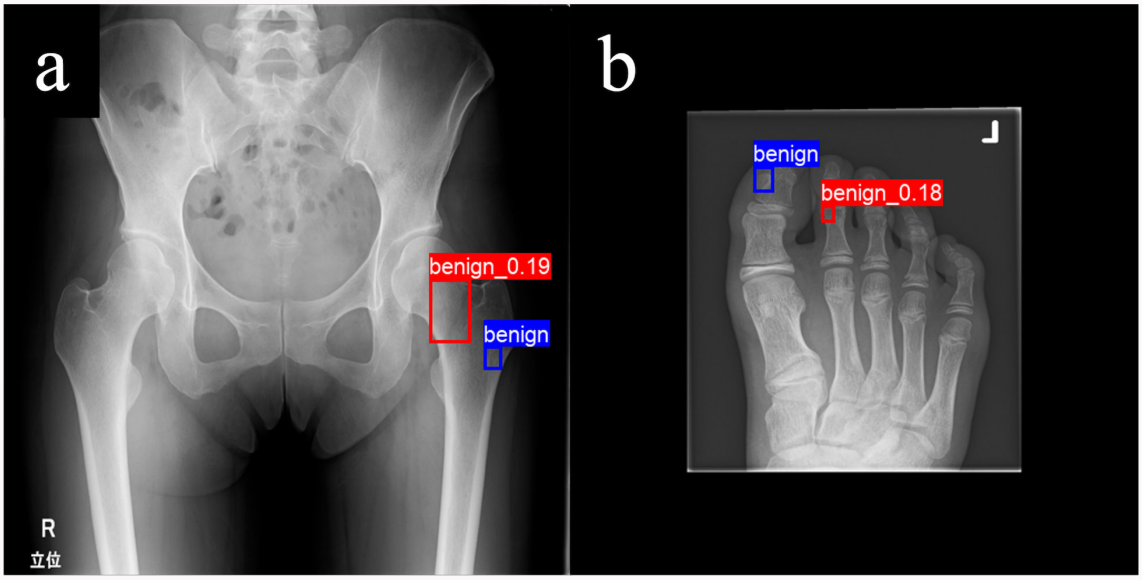

Fig. 3
Incorrect predictions by the DINO model. The blue bounding box is a ground-truth bounding box, and the red bounding box is a predicted bounding box. a) Osteoid osteoma case (32-year-old female), b) subungual exostosis case (13-year-old male). All radiographs were anteroposterior views obtained at the initial visit.

Results

Detection performance evaluations of DINO versus YOLO

Table II shows the tumour detection rates achieved by the object detection models. The DINO model showed better detection performance than the YOLO model (85.7% (95% CI 81.5 to 89.8) vs 80.1% (95% CI 77.2 to 82.9); p = 0.014, paired *t*-test). In particular, the DINO model outperformed the YOLO model in the detection rate of malignant tumours in all folds. Figure 2 shows the results of the successfully detected and classified tumours by the DINO model. Figure 3 illustrates examples of the unsuccessfully detected results predicted by the DINO model.

The object detection models had a mean detection rate for benign tumours that was inferior to that for malignant tumours. The bounding box area% of benign tumours was significantly smaller than that of malignant tumours ($p < 0.001$, Mann-Whitney U test). The term "bounding box area %" is the area ratio of the ground-truth bounding box of a tumour over the whole image.

Classification performance evaluations

Table III shows classification results for the DINO model, as a superior model, three orthopaedic oncologists, and three general orthopaedic surgeons. The classification performance of the DINO model was higher than those of general orthopaedic surgeons and comparable to those of oncologists. The DINO model showed significantly higher accuracy than Dr. GA and Dr. GC of general orthopaedic surgeons (77.0% vs 65.5% and 63.7%; p = 0.016 and 0.031). The sensitivity of the DINO model was significantly higher than that of Dr. SA of the orthopaedic oncologists and Dr. GA, Dr. GB, and Dr. GC of the general orthopaedic surgeons (84.3% vs 58.8%, 49.0%, 64.7%, and 62.7%; p = 0.003, < 0.001, 0.012, and 0.018, McNemar test). There were no statistically significant differences between the specificity of the DINO model and that of Dr. SA-C and Dr. GA-C (71.0% vs 83.9%, 82.3%, 74.2%, 79.0%, 74.2%, and 64.5%; p = 0.059, 0.109, > 0.999, 0.467, 0.827, and 0.414, McNemar test). The DINO scored the second highest in F-measure (76.8% for the DINO model and 79.3% for Dr. SC, an orthopaedic oncologist), surpassing all general orthopaedic surgeons.

In some challenging malignant tumours that were incorrectly classified by two or more doctors, the DINO model correctly classified such problematic cases, achieving a correct classification rate of 78.6% (22 out of 28 tumours) (Supplementary Table i). The DINO model also correctly classified 62.5% (5 out of 8) of the malignant tumours misclassified by all general orthopaedic surgeons. Furthermore, the DINO model correctly classified 75.0% (18 out of 24) of malignant tumours that were misclassified by one or more orthopaedic oncologists. In contrast, the DINO model was incorrect in only one of the cases, even though all the doctors made the correct malignant classification.

Discussion

In this study, we first developed deep learning models for bone tumour detection and classification of benign and malignant tumours using the Transformer-based object detection models. This is the first study to establish the Transformer-based model to detect and classify malignant/benign bone tumours and to compare the accuracy of different architecture-based object detection models. Due to the Transformer's ability to extract features by considering the relationships across the entire image, the DINO model had significantly higher detection rates than the YOLO-v8x model (85.7% vs 80.1%; p = 0.014). The DINO model obviously showed higher detection rates, especially for malignant tumours (89.0% vs 81.1%).

Notably, the DINO model's classification accuracy score surpassed those of all general orthopaedic surgeons. Misdiagnosis of malignant tumours is disadvantageous to patients, but the DINO model showed a significantly higher sensitivity than that of one orthopaedic oncologist and three

Table II. DINO and YOLO models' detection rates for all test sets.

Tumour type	DINO model						YOLO model					
	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	Mean	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	Mean
All	87.6% (113/129)	88.3% (113/128)	88.3% (113/128)	81.3% (104/128)	82.9% (107/129)	85.7% (550/642)	81.4% (105/129)	78.9% (101/128)	76.6% (98/128)	82.0% (105/128)	81.4% (105/129)	80.1% (514/642)
Benign	84.4% (65/77)	83.8% (62/74)	85.5% (65/76)	81.3% (61/75)	81.6% (62/76)	83.3% (315/378)	84.4% (65/77)	74.3% (55/74)	76.3% (58/76)	78.7% (59/75)	82.9% (63/76)	79.4% (300/378)
Malignant	92.3% (48/52)	94.4% (51/54)	92.3% (48/52)	81.1% (43/53)	85.0% (45/53)	89.0% (235/264)	76.9% (40/52)	85.2% (46/54)	76.9% (40/52)	86.8% (46/53)	79.2% (42/53)	81.1% (214/264)
Mean IoU	0.56	0.55	0.57	0.55	0.52	0.55	0.58	0.53	0.52	0.57	0.57	0.55

DINO, Detection with transformers with Improved deNoising anchOr boxes; IoU, Intersection over Union; YOLO, You Only Look Once.

Table III. Classification performance of the DINO model, three orthopaedic oncologists, and three general orthopaedic surgeons in the evaluation test set.

Evaluation metric	DINO	Dr. SA	Dr. SB	Dr. SC	Dr. GA	Dr. GB	Dr. GC
Accuracy	77.0% (87/113)	72.6% (82/113)	77.0% (87/113)	79.6% (90/113)	65.5% (74/113)	69.9% (79/113)	63.7% (72/113)
Sensitivity	84.3% (43/51)	58.8% (30/51)	70.6% (36/51)	86.3% (44/51)	49.0% (25/51)	64.7% (33/51)	62.7% (32/51)
Specificity	71.0% (44/62)	83.9% (52/62)	82.3% (51/62)	74.2% (46/62)	79.0% (49/62)	74.2% (46/62)	64.5% (40/62)
Precision	70.5% (43/61)	75.0% (30/40)	76.6% (36/47)	73.3% (44/60)	65.8% (25/38)	67.3% (33/49)	59.3% (32/54)
F-measure	76.8%	65.9%	73.5%	79.3%	56.2%	66.0%	61.0%

SA, SB, SC: orthopaedic oncology specialists; GA, GB, GC: general orthopaedic surgeons.
DINO, Detection with transformers with Improved deNoising anchOr boxes.

general orthopaedic surgeons. Additionally, the F-measure score of the DINO was the second highest among the model and all six doctors. Interestingly, the DINO model correctly classified some challenging cases that two or more doctors misclassified. Furthermore, only two cases were misclassified by the DINO, with all doctors getting them right. Thus, the DINO model may assist doctors in diagnosing tumours and reduce the risk of misclassification by them.

Although DINO demonstrated superior performance, we also investigated its limitations. The model's misclassifications primarily occurred in cases involving challenging tumours that were diagnostically difficult for orthopaedic oncologists or located in unusual sites. Of the seven malignant tumours misclassified by both DINO and at least one doctor, three were misdiagnosed by all three orthopaedic oncologists. Additionally, three tumours were located in atypical sites: giant cell tumours in the fibula and calcaneus and Ewing's sarcoma in the radius. These findings indicate the need for further research focusing on cases that are difficult to diagnose on radiographs and tumours occurring in atypical sites.

Few related studies have reported on deep-learning models for the fully automated detection and classification of benign and malignant bone tumours using full-field bone radiographs of various body parts.[18,19] Although the data used were different, we compared the accuracies reported in their studies with those obtained in our study and discussed the differences between their datasets and ours in the following section. A previous report on tumour detection and classification using the Detectron2 showed a tumour detection rate of 82.0% (91 of 111, IoU > 0.0), classification accuracy of 80.2% (89 out of 111 tumours), and classification sensitivity of 62.9% (22 out of 35 tumours).[18] Another report on tumour detection and classification using the YOLO-v5 model showed a high tumour detection rate and classification sensitivity of 74.6% (47/63 cases) when intermediate-type tumours were classified as malignant tumours.[19] However, the tumour types included in the reports were limited to 16 and 29. Regarding the latter report, approximately 43.9% (65/148 cases) of the benign tumours of test data were osteochondroma, which may have increased detection rates and classification accuracy. Although this study included 40 types of bone tumours, the mean detection rate was relatively high (85.7%). Furthermore, the sensitivity of the DINO model was 84.3%, indicating its high capability for classifying malignant tumours. Although bone tumours are rare, our sample size was comparable to those in related studies and is sufficient for conducting a meaningful analysis.

The DINO model is based on a Transformer and understands a whole image's local and global features. In addition, it has been shown that the prediction error of the Transformer is more consistent with that of humans than that of CNN.[27,40,41] Therefore, the DINO model may classify tumours more accurately than CNN-based models such as YOLO.

This study has some limitations. First, although this was unavoidable considering the rarity of bone tumours, 642 cases, the number of cases studied, needed to be

sufficiently large compared to studies on common cancers. Second, the input image size was limited to 512 × 512 pixels due to Graphics Processing Unit (GPU) memory capacity constraints. Future studies will require increased GPU memory capacity to process higher-resolution images. Third, the cases studied were limited to instances of limbs and did not include vertebrae, pelvis, or ribs. Evaluating these trunk bones is expected to be difficult using radiography alone, so additional studies using other imaging methods are needed. Fourth, the current model cannot detect and classify targets other than bone tumours. The models were not trained on images without tumour lesions or with fractures or degenerative diseases. Future studies should include those images to make clinical applications more feasible. Furthermore, it could be a future option to take into account clinical findings such as pain, sex, age and blood test results, in addition to imaging to classify malignant/benign bone tumours more precisely.

In summary, the Transformer-based DINO model automatically detects bone tumours in full-field limb radiographs more accurately than the CNN-based YOLO model. This model could serve as a valuable support tool in daily clinical practice, particularly assisting general orthopaedic surgeons who may be less familiar with bone tumours. While radiological review by doctors remains essential, especially for complex cases, the model's high-accuracy diagnostic capabilities could help reduce doctor workload through automated tumour detection and classification, particularly in avoiding misdiagnosing malignant tumours as benign ones. Future studies are warranted to make clinical applications more feasible.

Supplementary material

Text describing the deep learning object detection models, dataset, image preprocessing, and model's training used in the study. Table showing malignant tumours misclassified by one or more doctors in the evaluation test set. Figures showing overview of five-fold cross-validation and distribution of cases and images, and examples of image rotation and cropping for preprocessing. TRIPOD+AI checklist is also included.

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Data sharing

The data that support the findings for this study are available to other researchers from the corresponding author upon reasonable request.

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Ethical review statement

The study was approved by the Institutional Review Board of Kyushu University Hospital (No.22316-00, No.22089-00). Informed consent was obtained in the form of an opt-out on the website, and those who rejected it were excluded.

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